Identity-by-descent mapping for rare variants in biobank-scale datasets

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BACKGROUND

- We have implemented an identity-by-descent (IBD) approach to identify rare, causal variants for common diseases by creating an approximate genealogy of the Million Veterans Program (MVP) biobank.
- This IBD method is based on the observation that long-range haplotype sharing between study subjects necessitates inheritance from a recent common ancestor.
- Biobank genomic datasets contain hundreds of thousands of multi-generational extended pedigrees.
- Furthermore, IBD inference offers a more statistically powerful approach to identify rare variants contributing to disease than case-control association.

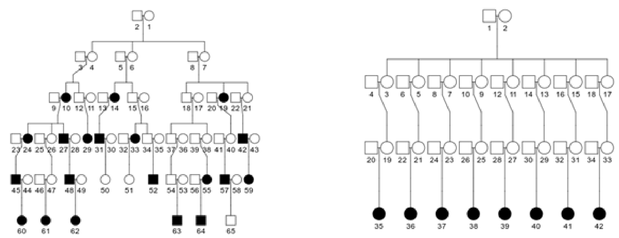
DATA RESOURCES

Million Veterans Program (MVP): ~650K subjects with SNP array data and ~102K with whole genome data

Penn Medicine Biobank (PMBB): ~45K subjects with SNP array data and whole exome data

UKBiobank (UKB): ~500K subjects with whole genome data[1]

AN APPROXIMATED GENEALOGY

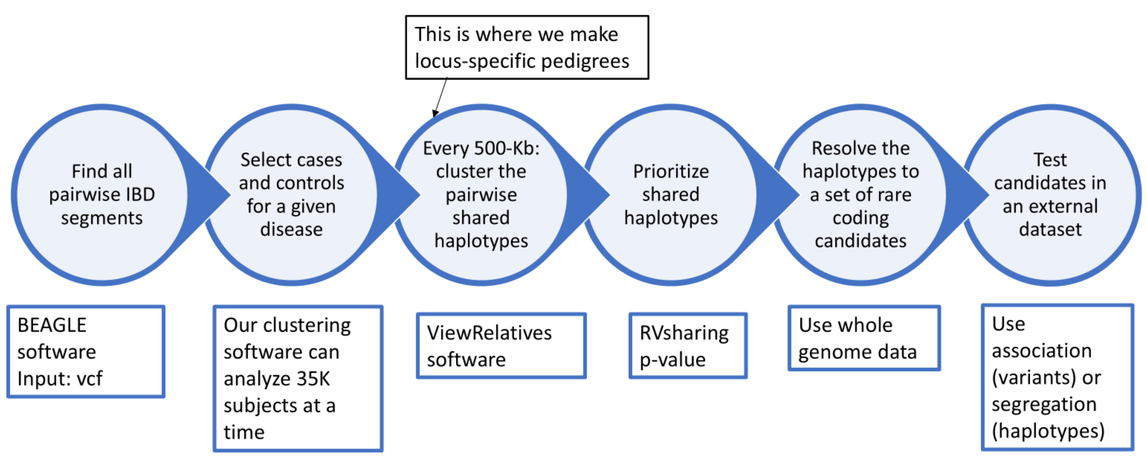


A *CDKN2A* pedigree and a *conservative* IBD-inferred pedigree

METHODS

- Within biobank datasets, pedigrees can be found by looking for patterns in the genetic marker data:
  - When people all share a long run of the same strand of DNA, they must have inherited it from the same common ancestor
  - These are referred to as 'identity-by-descent' (IBD) runs of shared genetic material, sometimes referred to as 'haplotypes'
- Pairwise IBD events are easy to find[2]
- Pairwise IBD events can be clustered into extended pedigrees using a graphical modeling algorithm
- Pedigrees can be analyzed via segregation techniques which can be much more powerful for rare variants than association
- 'Approximate' pedigrees found in MVP can be expanded into other biobank datasets for confirmation of linkage to a disease

How we create IBD-inferred indicator variables



RESULTS

PROOF OF CONCEPT: Amyloidosis

- Amyloidosis is a useful benchmark of our method because a subset of amyloidosis subjects in MVP are known to be explained by rare variants in the *TTR* gene (v142i; MAF=0.1%).
- 1008 MVP subjects had a diagnosis of amyloidosis.
- A nonsynonymous variant chr18:31595157:A:G in *TTR* gene had RVsharing $p = 1.3e^{-3}$ (CADD score=20.5)
- ClinvarID 28460: Pathogenic for 'familial amyloid neuropathy'

EXPLORATORY ANALYSIS: Cirrhosis

- Out of 161 candidate variants emerging from our analysis of MVP, the nonsynonymous variant chr17:29121027:G:A in *MYO18A* had RVSharing[3] $p=3.6e^{-4}$ in MVP (CADD score=26.2)
- This gene had a gene-level association[4] $p=3.1e^{-2}$ in PMBB and a gene-level association $p=3.2e^{-2}$ in UKB
- MYO18A encodes protein isoforms involved in pleiotropic function (e.g. cellular morphology, migration, immunity, viral infectivity, cancer).
- Taken together, these findings indicate that variants in *MYO18A* may convey risk for cirrhosis.

CONCLUSIONS

- IBD-based gene discovery methods offer a statistically powerful approach to identify rare variants in biobank datasets and strongly complements other approaches currently being applied.
- Variants in *MYO18A* may confer risk for cirrhosis

CONTACT

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References:

1. Genebass: https://app.genebass.org/
2. Browning BL, Browning SR. A fast, powerful method for detecting identity by descent. Am J Hum Genet. 2011;88:173-182.
3. Bureau A, Younkin SG, Parker MM, et al. Inferring rare disease risk variants based on exact probabilities of sharing by multiple affected relatives. Bioinformatics. 2014;30(15):2189-96.
4. Zhou W, Zhao Z, Nielsen JB, et al. Scalable generalized linear mixed model for region-based association tests in large biobanks and cohorts. Nat Genet. 2020;52(6):634-639.